BYOP Project Report

Python Morning Briefing Bot

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Course: Fundamentals of AI and ML

1. Introduction and Problem Statement

The Problem I Chose: In the modern digital age, university students face massive information overload the moment they wake up. Checking the weather, reading the news, finding daily inspiration, and planning the commute to campus or nearby cities requires opening at least four different mobile applications. This context-switching is not only time-consuming but often leads to "doom-scrolling" and morning distractions.

Why It Matters: Time management and digital well-being are critical for academic success. By consolidating essential morning information into a single, text-based terminal output, students can get the data they need to start their day without being sucked into the distraction loops of modern smartphone apps.

2. Approach to Solving the Problem

To solve this, I developed a Python-based **Morning Briefing Bot**. My approach was to build a modular command-line interface (CLI) tool that fetches live, dynamic data from various web sources and compiles it into a clean, readable dashboard.

The core of the project relies on **REST APIs** and **Python logic**. I broke the script down into distinct, single-purpose functions:

- get_weather(): Uses the Open-Meteo API to fetch exact temperature and wind data for the VIT Bhopal campus coordinates.
- get_bhopal_news(): Uses the NewsAPI to search specifically for articles related to "Bhopal."
- get_quote(): Uses the ZenQuotes API to fetch a randomized daily inspirational quote.
- get_traffic_route(): Uses algorithmic logic to simulate commute times and evaluate the best route to take for the day.

3. Key Decisions Made

Throughout the development process, I had to make several architectural decisions to keep the project efficient and scoped appropriately:

- **API Selection:** I specifically chose the *Open-Meteo API* for weather because it does not require an API key, reducing the overhead of managing multiple secret keys.
- **Dynamic News Sourcing:** Initially, the bot just grabbed the top 3 most recent articles. However, this meant seeing the same headlines every time the script was run. I decided to pull a larger pool of articles and use Python's random.sample() function to select 5 random articles. This ensures the output feels fresh and dynamic upon every execution.

- **Traffic Logic Simulator vs. Paid API:** I originally wanted to use the Google Maps API for live traffic data. However, since that requires a paid billing account, I decided to engineer a "Traffic Logic Simulator." By using Python's random module to generate travel times and conditional statements (if/elif/else) to determine traffic severity, I was able to demonstrate my algorithmic problem-solving skills without incurring financial costs.

4. Challenges Faced

- **Handling Unstructured JSON Data:** Extracting the exact data points (like nested temperatures or article titles) from massive JSON responses was initially tricky. I had to learn how to properly navigate nested dictionaries and lists in Python.
- **Error Handling:** Early versions of the script would crash completely if an API server went down or if my Wi-Fi disconnected. I overcame this by implementing try/except blocks and response.raise_for_status(), ensuring the bot gracefully returns an error message for a specific module rather than breaking the whole program.
- **API Key Security:** I learned that hardcoding API keys directly into public GitHub repositories is a major security risk, as web scrapers can steal them. Managing this vulnerability was a key learning curve in understanding professional deployment practices.

5. What I Learned

Through this Bring Your Own Project capstone, I significantly improved my practical Python skills. Specifically, I learned:

a. How to make HTTP requests using the requests library.

b. How to parse and manipulate JSON data into readable string formats.

c. How to use the datetime module to stamp script executions.

d. The importance of modular coding—keeping separate tasks in separate functions makes debugging incredibly easy.

Ultimately, this project pushed me beyond theoretical textbook exercises and allowed me to build a practical software tool that I can actually use in my daily life at VIT Bhopal.

6. Running Code Screenshot

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 Good Morning! Here is your VIT Bhopal briefing.
 Generated on: Monday, March 30, 2026 at 11:42:23 PM
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🌟 "Great things are not accomplished by those who yield to trends and fads and popular opinion." - Jack Kerouac
🌤️ VIT BHOPAL WEATHER:
  🌡️Temperature: 25.6°C | 💨 Wind: 3.4 km/h

🗺️ COMMUTE & ROUTE SELECTION:
  🚗 Route A (Indore-Bhopal Highway): 39 mins (🟢 Light Traffic)
  🚗 Route B (Local Village Road): 53 mins (🔴 Heavy Traffic)

  ✅ Recommendation: Take Route A (Indore-Bhopal Highway) today!

📰 LATEST BHOPAL NEWS:
  1. What a mess! 'Lizard' found in meal at RGPV, staff terms it capsicum & eats it
  2. What do I tell them? Gas agencies avoid calls as eateries ring shortage alarm in Bhopal amid Middle East conflict
  3. Tension in Bhopal after 'cattle remains' found near temple; right-wing organisations protest, block traffic
  4. Don't want third term as Rajya Sabha member: Digvijaya Singh; rejects retirement plan rumours
  5. LPG crisis: Madhya Pradesh traders body seek limited gas supply to prevent shutdown of hotels, eateries
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